Global Justice Situator

A webpage that shows users how their income evolves under the **Sustainable Convergence (SC)** scenario from Chancel et al. (2026) and Bothe et al. (2026), part of the Global Justice Report.

Language and tooling

- **Data processing**: R (readxl). Entry point: `code_simulator/prepare_data.R`
- **Webpage**: HTML / CSS / JavaScript (vanilla)
- **Local testing**: XAMPP on Windows 10

Key concepts

Term	Definition
gpercentile	Within-country income percentile group, represented by its lower-bound percentage (0, 1, ..., 98 for standard groups; 99, 99.1, ..., 99.999 for fine top-1% groups)
SC scenario	Sustainable Convergence: global policies reduce inequality to k=5 (T10/B10 ≈ 5) by 2100; world average income converges to ~60,000 EUR PPP 2025
GIT	Global Income Transfers — international lump-sum per-capita redistribution mechanism in the SC scenario
posttax income	After domestic taxes and transfers; "income" in this project also adds GIT (post-GIT)
EUR PPP 2025	All monetary values in constant 2025 purchasing-power-parity euros
k value	Inequality parameter: k = T10/B10 ratio; k=5 is the SC 2100 target

Data sources

Fetched 2026-05-24:

- **Chancel et al. (2026)** — macro scenarios: `data/Chancel/Chanceletal2026Appendix_MacroScenarios.xlsx`
- **Bothe et al. (2026)** — within-country distributions: `data/Bothe/Botheetal2026AppendixDistribution.xlsx`

Key sheets in Botheetal2026AppendixDistribution.xlsx

Sheet	Content
P1e	2025 average posttax income by gpercentile group (127 groups × 67 countries). Values in EUR PPP 2025/year. Used as starting distribution.
I2i	SC scenario: average posttax net income time series 1800–2100 (years × 67 countries).

Sheet	Content
I1a	SC scenario: T10 income share time series.
I1b	SC scenario: B50 income share time series.
I1c	SC scenario: T1 income share time series.
I1e	SC scenario: B10 income share time series.
I1h	SC scenario: M40 income share time series.
I5e	SC scenario: P10 income threshold as ratio to average income (pre-GIT, time series).
I5b	SC scenario: P50 income threshold as ratio to average income (pre-GIT, time series).
I5c	SC scenario: P99 income threshold as ratio to average income (pre-GIT, time series).
I5d	SC scenario: P99.9 threshold as ratio to average income (pre-GIT).
I5f	SC scenario: P99.99 threshold as ratio to average income (pre-GIT).
I5g	SC scenario: P99.999 threshold as ratio to average income (pre-GIT).
I9i	SC scenario: average posttax net income (post-GIT, time series).
I9e	SC scenario: B10 average income (post-GIT, time series). Bracket [0, 0.10].
I9b	SC scenario: B50 average income (post-GIT). Bracket [0, 0.50].
I9h	SC scenario: M40 average income (post-GIT). Bracket [0.50, 0.90].
I9a	SC scenario: T10 average income (post-GIT). Bracket [0.90, 1.00].
I9c	SC scenario: T1 average income (post-GIT). Bracket [0.99, 1.00].
I9d	SC scenario: top 0.1% average income (post-GIT). Bracket [0.999, 1.00].
I9f	SC scenario: top 0.01% average income (post-GIT). Bracket [0.9999, 1.00].
I9g	SC scenario: top 0.001% average income (post-GIT). Bracket [0.99999, 1.00].
K1a/K2a	SC scenario: T10 wealth share / average (time series).
K1b/K2b	B50 wealth share / average.
K1c/K2c	T1 wealth share / average.
K1e/K2e	B10 wealth share / average.
K1h/K2h	M40 wealth share / average.
H1	SC 2100 target distribution shape (k=5): B50=37.7%, M40=44.1%, T10=18.1%, T1=2.6%.
P1a/P1b	Target distribution shapes at various k values (k=1..100).
H3b	Global 2025 income and wealth distribution by global percentile (for reference).

I9 series (I9a–I9i, all post-GIT group averages) are used as anchors for income.csv. I5 (pre-GIT threshold ratios) is used only for income_pre_git.csv. I10 contains post-GIT inequality ratios (T10/B50 etc.), not group averages, so it is not used.

Generated data files

`data/income_pre_git.csv`

Posttax income thresholds at 6 key percentiles, **before** Global Income Transfers, for each country and each year 2020–2100.

Column	Description
`country`	Country/region code (66 values incl. "World")
`year`	Year, 2020–2100
`p10`	P10 threshold = I5e × I2i (EUR PPP 2025/year)
`p50`	P50 threshold = I5b × I2i
`p99`	P99 threshold = I5c × I2i
`p99.9`	P99.9 threshold = I5d × I2i
`p99.99`	P99.99 threshold = I5f × I2i
`p99.999`	P99.999 threshold = I5g × I2i

Dimensions: 66 countries × 81 years = 5,346 rows.

`data/income.csv`

SC scenario posttax **post-GIT** income, all 127 gpercentile groups, all years 2020–2100.

Column	Description
`country`	Country/region code
`gpercentile`	Lower bound (%) of the percentile bracket: 0, 1, ..., 98 (standard groups, 1 pp wide), then 99, 99.1, ..., 99.8 (0.1 pp), 99.9, 99.91, ..., 99.98 (0.01 pp), 99.99, ..., 99.999 (0.001 pp)
`income_2020` ... `income_2100`	Income in EUR PPP 2025/year for each year

Methodology:

1. **2025 base**: all 127 groups taken directly from P1e (EUR PPP 2025).
2. **Other years**: for each year t ≠ 2025, compute a **scale factor per group** as `group_avg_t / group_avg_2025`. The 2025 group average is derived from P1e (simple mean of equal-width rows within each group); the year-t group average is derived from I9 post-GIT series. All rows within a group are multiplied by the same scale factor, preserving the within-group relative income differences from P1e (consistent with the Type II Pareto intra-group functional form described in Appendix A of Bothe et al.).

Rows	Group	Bracket	I9 source (post-GIT)
1–10	B10	[0, 0.10]	I9e (B10 avg)
11–50	p10_50	[0.10, 0.50]	(I9b×0.5 − I9e×0.1) / 0.4
51–90	M40	[0.50, 0.90]	I9h (M40 avg)
91–99	p90_99	[0.90, 0.99]	(I9a×0.1 − I9c×0.01) / 0.09
100–108	p99_999	[0.99, 0.999]	(I9c×0.01 − I9d×0.001) / 0.009
109–117	p999_9999	[0.999, 0.9999]	(I9d×0.001 − I9f×0.0001) / 0.0009
118–126	p9999_99999	[0.9999, 0.99999]	(I9f×0.0001 − I9g×0.00001) / 0.00009
127	top0001	[0.99999, 1.00]	I9g (top 0.001% avg)

All anchors are post-GIT (I9 series). No pre-GIT I5 threshold ratios are used for income.csv.

Dimensions: 66 countries × 127 groups = 8,382 rows × 83 columns.

`data/income_world.csv`

World income distribution under SC at 6 target years (2025, 2030, 2035, 2050, 2080, 2100).

Column	Description
`gpercentile`	Global income percentile rank, 1–100
`income_2025` ... `income_2100`	Income in EUR PPP 2025/year

Dimensions: 100 rows × 7 columns. Used for the distribution chart on the webpage.

`data/wealth.csv`

SC scenario wealth, all 100 gpercentiles, target years 2025–2100.

Column	Description
`country`	Country/region code
`gpercentile`	Within-country wealth percentile rank, 1–100
`wealth_2025` ... `wealth_2100`	Wealth in EUR PPP 2025

Methodology: 5-group scaling (B10, p10-50, M40, T10-T1, T1) using K2 group average series, starting from K2 group averages at 2025.

Dimensions: 6,600 rows × 8 columns.

`data/wealth_world.csv`

World wealth distribution at 6 target years.

Dimensions: 100 rows × 7 columns.

Countries and regions

66 values total, including "World":

Individual countries: AE, AR, AU, BD, BR, CA, CD, CI, CL, CN, CO, DE, DK, DZ, EG, ES, ET, FR, GB, ID, IN, IR, IT, JP, KE, KR, MA, ML, MM, MX, NE, NG, NL, NO, NZ, PH, PK, RU, RW, SA, SD, SE, TH, TR, TW, US, VN, ZA

Residual "other" groups (WID.world codes): OA (Other Russia/Central Asia), OB (Other East Asia), OC (Other Western Europe), OD (Other Latin America), OE (Other MENA), OH (Other North America/Oceania), OI (Other South & SE Asia), OJ (Other Sub-Saharan Africa), QM (Eastern Europe)

Aggregate regions: East Asia, Europe, Latin America, Middle East North Africa, North America Oceania, Russia Central Asia, South & South-East Asia, Sub-Saharan Africa

Plus: World

Running the data pipeline

```
# in R, from the project root:
Rscript --vanilla code_simulator/prepare_data.R  # run from project root
```

Outputs all five CSV files to `data/`. Runtime: ~2 minutes.

Local testing

Place the project folder under XAMPP's `htdocs/` and access via `http://localhost/global_justice/`.